March 23, 2026

Kenshi KUROKI 黒木 健志

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Personal Data

Date of birth:	Jan. 21, 1996
Place of birth:	Nagasaki, Japan
Nationality:	Japan

Present Position and Affiliation

Postdoctoral Fellow
Quark Matter Research Center
Institute of Modern Physics, Chinese Academy of Sciences, China

Academic Position

Jul. 2025–Present	**Postdoctoral Fellow** Institute of Modern Physics Chinese Academy of Sciences, China
Jul. 2025–Present	**Co-Researcher Partnership** Faculty of Science and Technology Sophia University, Japan
Apr. 2024–Jun. 2025	**Postdoctoral Fellow** Faculty of Science and Technology Sophia University, Japan
Apr. 2021–Mar. 2024	**Research Assistant** Faculty of Science and Technology Sophia University, Japan

Education

Apr. 2021–Mar. 2024	**Doctor of Philosophy in Physics (Ph.D.)** Department of Physics, Faculty of Science and Technology Sophia University, Japan Supervisor: Prof. Tetsufumi Hirano Dissertation title: "Hadron correlation and interaction from a dynamical model in high-energy nuclear collisions"
Apr. 2019–Mar. 2021	**Master of Science in Physics (M.Sc.)** Department of Physics, Faculty of Science and Technology Sophia University, Japan Supervisor: Prof. Tetsufumi Hirano Dissertation title: "Effects of event-by-event fluctuations in ultra-central heavy-ion collisions"

Apr. 2015–Mar. 2019	**Bachelor of Science (B.S.)** Department of Engineering and Applied Science, Faculty of Science and Technology Sophia University, Japan

Award

Jun. 2024	**Exemption from Return for Particularly Outstanding Achievement**, Japan Student Services Organization (JASSO)

Fellowship/Scholarship

Apr. 2021–Mar. 2024	Sophia Alumni Entrepreneurs Club Scholarship
Apr. 2021–Mar. 2024	Japan Student Services Organization (JASSO) Scholarship
Apr. 2021–Mar. 2024	Sophia University Graduate School Scholarship for Fostering Researchers in Doctoral Program

Membership of Academic Society

The Physical Society of Japan

Community Service

Organizer for	Jet Modification and Hard-Soft Correlations (SoftJet 2024), Tokyo, Japan, Sep. 28–29, 2024, (Local Organizing Committee)

Computational Skill

Operating System	Linux, Windows
Programming	C/C++, Python, Wolfram
Miscellaneous	Shell (Bash, zsh), LaTeX, gnuplot, Github, Microsoft Office

Language Skill

Japanese	Native
English	Proficient

Reference

Shunzo Kumano	Professor, Quark Matter Research Center Institute of Modern Physics Chinese Academy of Sciences	No.1 Xinqiao North Road, Henan Bei'an, Huicheng District, Huizhou, Guangdong, 516000, China kumano@impcas.ac.cn
Tetsufumi Hirano	Professor, Department of Physics, Faculty of Science and Technology, Sophia University	7–1, Kioi-cho, Chiyoda-ku, Tokyo, 102–8554, Japan +81 3 3238 3434 hirano@sophia.ac.jp

Publications

Citation data is based on INSPIRE (Mar. 23, 2026).

Refereed Journal

- **S. Kumano and K. Kuroki**, (Corresponding author)

  "Tensor-polarized twist-3 parton distribution functions $f_{LT}(x)$ for the spin-1 deuteron by using twist-2 relations", **Phys. Lett. B 875, 140348 (2026)**, arXiv:2509.05046 [hep-ph]. ***[3 citations]***

- **K. Kuroki, A. Sakai, K. Murase, and T. Hirano**, (Corresponding author)

  "Hydrodynamic fluctuations and ultra-central flow puzzle in heavy-ion collisions", **Phys. Lett. B 842, 137958 (2023)**, arXiv:2305.01977 [nucl-th]. ***[17 citations]***

Conference Proceedings

- **K. Kuroki and T. Hirano**, (Corresponding author, Refereed)

  "p–ϕ femtoscopic correlation analysis using a dynamical model", **The 21st International Conference on Strangeness in Quark Matter (SQM 2024)**, Strasbourg, France, Jun. 5, 2024, **EPJ Web Conf. 316, 03009 (2025)**, arXiv:2410.01204 [hep-ph]. ***[2 citations]***

Presentations

The speaker is marked with a circle.

Invited Talk

- ◯**K. Kuroki**,

  "Effects of collision dynamics on pϕ correlation function" (Japanese), **2nd Workshop on Intersection of J-PARC and Heavy Ion Collision Experiments**, Tokai, Ibaraki, Japan, Mar. 7, 2025.

- ◯**K. Kuroki**,

  "Effects of dynamics on interaction study via femtoscopy" (Japanese), **Go-Forward 2025**, Nagasaki, Japan, Feb. 28, 2025.

- ◯**K. Kuroki**,

  "Effects of collision dynamics on pϕ femtoscopy", **10th Asian Triangle Heavy-Ion Conference - ATHIC 2025**, Berhampur, India, Jan. 15, 2025.

- ◯**K. Kuroki**,

  "p–ϕ interaction from femtoscopy using a dynamical model", **International workshop on J-PARC hadron physics 2024 (J-PARC Hadron 2024)**, Tokai, Ibaraki, Japan, Jul. 23, 2024.

- ◯**K. Kuroki**,

  "Effects of hydrodynamic fluctuations on anisotropic flow in ultra-central heavy-ion collisions" (Japanese), **Post Quark Matter 2019**, Nagoya University, Nagoya, Japan, Dec. 22, 2019.

Invited Seminar

- ◯**K. Kuroki**,

  "Study on hadron correlations and interactions using a dynamical model" (Japanese), **42nd Heavy Ion Pub workshop**, Kyoto University, Kyoto, Japan, Mar. 26, 2025.

- ◯**K. Kuroki**,

  "Phenomenology of high-energy nuclear collisions using relativistic hydrodynamic models" (Japanese), **Sohaken Seminar**, Hiroshima University, Hiroshima, Japan, Jul. 3, 2023.

Contributed Talk

- ◯**K. Kuroki and S. Kumano**,

  "wist-3 tensor polarized distribution function f_{LT} of deuteron" (Japanese), **The Physical Society of Japan 2026 Spring Meeting**, Online, Japan, Mar. 23, 2026.

- ◯**A. Shigeyoshi, K. Kuroki, and T. Hirano**,

  "Bayesian analysis of QGP formation based on core-corona picture in high-energy proton-proton collisions" (Japanese), **The Physical Society of Japan 2026 Spring Meeting**, Online, Japan, Mar. 23, 2026.

- ◯**K. Kuroki and T. Hirano**,

  "p–ϕ femtoscopic correlation analysis using a dynamical model", **The 21st International Conference on Strangeness in Quark Matter (SQM 2024)**, Strasbourg, France, Jun. 5, 2024.

- ◯**K. Kuroki and T. Hirano**,

  "p–ϕ femtoscopy using a dynamical model" (Japanese), **The Physical Society of Japan 2024 Spring Meeting**, Online, Japan, Mar. 19, 2024.

- ◯**K. Kuroki, A. Sakai, K. Murase, and T. Hirano**,

  "Effects of hydrodynamic fluctuations in ultra-central heavy-ion collisions" (Japanese), **The Physical Society of Japan the 77th Annual Meeting**, Online, Japan, Mar. 15, 2022.

- ◯**K. Kuroki, A. Sakai, K. Murase, and T. Hirano**,

  "Effects of hydrodynamic fluctuations in ultra-central high-energy heavy-ion collisions" (Japanese), **The Physical Society of Japan the 75th Annual Meeting**, Online, Japan, Mar. 16, 2020.

Poster Presentation

- ◯**K. Kuroki and T. Hirano**,

  "p–ϕ correlation and interaction using a dynamical model" (Japanese), **Tutorial workshop for high-energy heavy-ion collision physics 2024**, Osaka University, Osaka, Japan, Aug. 6, 2024.

- ◯**K. Kuroki, A. Sakai, K. Murase, and T. Hirano**,

  "Effects of hydrodynamic fluctuations in ultra-central Pb-Pb collisions at LHC", **The 29th International Conference on Ultra-relativistic Nucleus-Nucleus Collisions (QM 2022)**, Online, Apr. 6, 2022.

- ◯**K. Kuroki, A. Sakai, K. Murase, and T. Hirano**,

  "Effects of hydrodynamic fluctuations on azimuthal flow in ultra-central heavy ion collisions", **The 28th International Conference on Ultra-relativistic Nucleus-Nucleus Collisions (QM 2019)**, Wuhan, China, Nov. 4, 2019.